Yulun Wu

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Education

University of Illinois Urbana-Champaign — **Aug. 2025 – Present**
Ph.D. in Information Sciences — *Champaign, IL, USA*
- Advisor: Prof. Yaoyao Liu

Tsinghua University — **Sept. 2022 – Jun. 2025**
M.Eng. in Software Engineering — *Beijing, China*

Tsinghua University — **Aug. 2018 – Jun. 2022**
B.Eng. in Automation — *Beijing, China*

Publications [Google Scholar] (*Equal contribution. ✉Corresponding author.)

1. **Yulun Wu**, Ruyi Zha, Wei Cao, Yingying Li, Yuanhao Cai✉, Yaoyao Liu.
Active View Selection with Perturbed Gaussian Ensemble for Tomographic Reconstruction. [PDF]
European Conference on Computer Vision (ECCV), 2026.

2. **Yulun Wu***, Han Huang*, Wenyuan Zhang, Chao Deng, Ge Gao✉, Ming Gu, Yu-Shen Liu.
Sparis: Neural Implicit Surface Reconstruction of Indoor Scenes from Sparse Views. [PDF]
Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2025. (Oral Presentation)

3. Han Huang*, **Yulun Wu***, Chao Deng, Ge Gao✉, Ming Gu, Yu-Shen Liu.
FatesGS: Fast and Accurate Surface Reconstruction with Gaussian Splatting from Sparse Views. [PDF]
Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2025. (Oral Presentation)

4. Han Huang, **Yulun Wu**, Junsheng Zhou, Ge Gao✉, Ming Gu, Yu-Shen Liu.
NeuSurf: On-Surface Priors for Neural Surface Reconstruction from Sparse Input Views. [PDF]
Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), 2024.

5. Wei Cao, Hao Zhang, Fengrui Tian, **Yulun Wu**, Yingying Li, Shenlong Wang, Ning Yu, Yaoyao Liu.
FreeOrbit4D: Training-Free Arbitrary Camera Redirection for Monocular Videos via Geometry-Complete 4D Reconstruction. [PDF]
ACM SIGGRAPH Conference Papers, 2026.
CVPR 2026 GCV Workshop, 2026. (Best Paper Award)
CVPR 2026 4D Vision Workshop, 2026. (Oral Presentation)

6. Chao Deng, Han Huang, Ziyi Ke, **Yulun Wu**, Ge Gao✉, Ming Gu.
BIMFusion: Texture Your Building Information Models Using Diffusion Generative Method. [PDF]
Journal of Computing in Civil Engineering, 2025.

Preprints

1. Wei Cao, Hao Zhang, Jiapeng Tang, **Yulun Wu**, Yingying Li, Ning Yu, Shenlong Wang, Yaoyao Liu.
Redirect4D-Bench: A Scalable Benchmark for Camera Redirection of Monocular Dynamic Videos with Pseudo-4D Ground Truth. [PDF]
Preprint, 2026.

Service

Conference Reviewer: ECCV (2026), CVPR (2026), ICME (2025–2026), ICASSP (2026), etc.

Journal Reviewer: IJCV.

Area Chair: CVPR 2026 HiGen Workshop.

Teaching Assistant: UIUC CS 598 Foundations of Data Curation (Fall 2026).

Work Experience

Intelligent Creation Group, ByteDance Inc. **May 2026 – Aug. 2026**
Research Intern *San Jose, CA, USA*

- Research Orientation: Controllable human-centric audio-visual generation and joint editing.

BIM Research Group, Tsinghua University **Jul. 2022 – Jun. 2025**
Research Assistant *Bejing, China*

- Research Orientation: 3D computer vision and graphics for architecture and construction.

Base Model Group, Beijing SenseTime Technology Co., Ltd. **Jul. 2021 – Apr. 2022**
Research Intern *Bejing, China*

- I worked as a vision algorithm researcher, mainly focusing on face recognition algorithms research.
- Our team won the 1st place in Masked Face Recognition Challenge & Workshop (Glint360k Track), ICCV 2021.

Honors & Awards

Best Paper Award, 4th Workshop on Generative Models for Computer Vision, CVPR, 2026.

Oral Presentation, 2nd 4D Vision Workshop, CVPR, 2026.

Oral Presentation, AAAI, 2025. (Top 4.6% of all submitted papers)

Tsinghua Comprehensive Excellence Scholarship, Tsinghua University, 2023, 2024.

Tsinghua Innovation Excellence Scholarship, Tsinghua University, 2021.

1st Place Award, Mased Face Recognition Challenge & Workshop, ICCV, 2021.